

Online Judge System

Development Chronicle & Architecture Analysis

1. Project Overview

An online code judging system built with FastAPI + SQLAlchemy async + PostgreSQL + Redis + ARQ. Supports Python, C++, and Java judging in Docker sandboxes. Features a multi-page SPA frontend with i18n, WebSocket live status, 101 classic problems with official solutions, contest/leaderboard system, and production-grade security (rate limiting, CSRF, audit logs, brute-force lockout).

Total: 20 commits over 3 development phases. ~2,000 lines backend Python, ~2,500 lines frontend JS, ~1,000 lines CSS, 8 HTML pages, 69 tests at 83% coverage.

2. Development Phases

Phase 1: Foundation (Commits: ed037f5 - a8f7d06)

Established the core architecture: FastAPI routes, SQLAlchemy async models, Docker Compose services, and the initial frontend portal.

Commit	Title	Achievement	Gaps at this point
ed037f5	Initialize online judge system	First commit: FastAPI app, 5 DB models, judge routing with subroutes, Docker Compose	Initial gaps: no pagination, no translations.
fe2adfd	Add multipage frontend portal	SPA with auth, problems, submit, submissions, admin pages only	Initial gaps: no pagination, no translations.
a8f7d06	Production-grade online judge system	Docker sandbox for code execution, rate limiting, audit logs, contest leaderboard, Prometheus	Initial gaps: no pagination, no translations.

Phase 2: Contest System & Multi-language (Commits: a41023b - f9e1128)

Added comprehensive contest/leaderboard system, C++/Java support, and extensive test coverage.

Commit	Title	Achievement	Notes
a41023b	Add service-layer and core-utils test suites	+9 tests covering all 5 services, Redis pool lifecycle, pagination edge cases	
f6c7397	Fix ruff E712 (== True)	Boolean comparison style fix for CI lint	-
f9e1128	Complete contest system	Contest model + API + frontend + leaderboard	Contest problem display NameError, CSRF page lock
8da3131	Fix test language to unsupported value	Updated test for multi-language schema change	
8dc36ec	Fix contest problems display + app stability	Reorder Pydantic schemas to fix NameError, remove --reload flag, add Contest nav links	

Phase 3: Security & Production Hardening (Commits: 4195564 - 645ddbc)

Addressed all production gaps: account lockout, CSRF protection, password change, alembic migrations, K8s deployment, comment system, CodeMirror language switching. CI went from 20 failing tests to 0.

Commit	Title	Achievement	Notes
4195564	Round 1 fixes	Contest retry dedup, registration start-time cap	CSRF middleware blocked all non-GET requests
7578566	Round 2 features	E2E tests (16-step flow), CodeMirror multi-lang	only 20 test failures due to CSRF 403s
efe1e0b	Search slug matching	Problem search now matches title AND slug-fields	
f59ac17/9ae10ed/25824bf	CSRF hardening (3 commits)	CSRF exempts Bearer auth, E2E self-contained	fixed testing: e2e solution_and_csrf needed inline problem
f017cc0	Ruff lint fixes	57 auto-fixable issues handled via --fix in CI	,-B9
8358837	CodeMirror C++/Java highlighting	Pre-load lang-cpp/lang-java, Compartment-based run	
73d002c	Admin solution load + contest marking	Load button fetches existing solution, submit page	
645ddbc	K8s Helm chart + comments	8-file Helm chart (deploy/svc/ingress/statefulset)	

3. Architecture

Layer: Frontend (JS ES modules) -> FastAPI Router -> Service Layer -> SQLAlchemy Models -> PostgreSQL

Async Judge: Router -> Redis/ARQ Queue -> Worker -> Docker Sandbox (Python/C++/Java)

9 DB tables: users, problems, test_cases, submissions, submission_test_results, audit_logs, contests, contest_problems, contest_participants, contest_submission_attempts, comments

Current Scores



4. Key Achievements

- 101 classic problems with official solutions (100% coverage)
- Multi-language judging: Python, C++, Java in Docker sandboxes
- Contest system: registration, scoring with penalty, freeze board, countdown, leaderboard
- 69 tests, 83% code coverage, CI enforces 75% minimum
- Security: rate limiting, CSRF tokens, httpOnly cookies, brute-force lockout, audit logs, soft delete
- Observability: structured JSON logging, Prometheus metrics, request ID tracking
- Frontend: 8-page SPA, CodeMirror editor with syntax highlighting, zh/en i18n, light theme, PWA
- DevOps: Docker Compose with health checks, GitHub Actions CI/CD, K8s Helm chart

5. Remaining Gaps & Future Work

- Frontend build step: 15 ES module imports -> single bundle with esbuild
- E2E browser tests with Playwright (beyond API-level E2E)
- CDN independence: bundle CodeMirror locally instead of esm.sh
- Redis high availability: sentinel/cluster configuration
- Horizontal scaling: docker.sock mount prevents multi-host workers
- Admin UI polish: batch operations, problem import/export

6. Project Statistics

Metric	Value
Total Commits	20
Backend Python Files	45+
Frontend JS Files	15
HTML Pages	9
Database Tables	11
API Endpoints	25+
Test Count	69

Online Judge System - Development Chronicle

Code Coverage	83%
Problems	101
Solutions	101 (100%)
Lines of Backend Code	~2,000
Lines of Frontend Code	~2,500
Lines of CSS	~1,000

Generated from git history of ZuoXing-0504/online-judge-system - 2026-05-08